

Quantum-Enhanced Agrivoltaic Digital Twin: Spectral Co-Design, Multifunctional Net Energy Benefit, and Quantum-Secured Telemetry

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Abstract

Agrivoltaic deployment is constrained by a system-level trade-off: static semi-transparent photovoltaic modules resolve the food–energy land conflict only by uniformly sacrificing either electrical output or crop productivity. Here we present a quantum-guided agrivoltaic system in which a semi-transparent organic photovoltaic layer couples to a plasmonic nanoparticle-on-mirror (NPoM) cavity acting as an active spectral shaper. By transmitting narrow bands at 750 nm and 820 nm, the system prepares polaron-dressed states in the Fenna–Matthews–Olson photosynthetic complex, extending exciton coherence by 50 % and achieving an area-weighted global canopy yield of 97.1 %. The NPoM sentinel layer enables multiplexed SERS and quantum dot diagnostics of agricultural contaminants and crop stress. On the macroscopic scale, the shield reduces crop evapotranspiration by 28 % and yields a Net Ecological Benefit of $72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$. A BB84 quantum link secures the IoT telemetry. Finally, techno-economic analysis of a Cameroonian micro-module gives a payback period of 4.32 yr, identifying conditions under which the multifunctional net benefit stays positive for the modeled 500 m^2 cooperative case.

Keywords: quantum agrivoltaics, FMO complex, vibronic coherence, plasmonic nanoparticle-on-mirror cavity, SERS diagnostics, FAO-56 evapotranspiration, agrivoltaic techno-economics, quantum digital twin

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1. Introduction

Agrivoltaics co-locates photovoltaic (PV) systems and agricultural cultivation to mitigate the land-use competition between food production and renewable energy generation [1, 2]. Land-use optimization and mobile-PV foundations for this field were established in *Applied Energy* [3, 4]. Recent syntheses systematize overhead, vertical, and tracking typologies with land-equivalent-ratio and levelized-cost metrics [5]. Across these studies, three gaps recur and remain unresolved. The optical architecture is *static*: semi-transparent PV claddings act as fixed spectral filters that attenuate irradiance uniformly, trading electrical output for crop biomass without optimization of the spectral light-sharing window [6, 7]. Dynamic, physics-guided spectral management has not been transposed into an operating agrivoltaic stack. The cost of stasis is quantifiable: in the mode-volume scan reported below, the forward transfer yield varies by a factor of 1.6 and the SERS enhancement by a factor of 48 across the accessible range (Table 2), so a statically chosen cladding fixes the energy–diagnostics balance at deployment with no ability to re-trade it. The scan probes the strong-coupling penalty only: all seven volumes carry the NPoM channel, and the filtered-excitation gains quantified next apply to the NPoM-free fraction at every point of the scan. Techno-economic and lifecycle assessments remain decoupled from the underlying device physics: electricity, crop, water, and carbon accounts are evaluated subsystem by subsystem, so multifunctional trade-offs are described rather than co-optimized [8, 9]. The cost of this decoupling is concrete: accounting that ignores the local transport physics would credit the canopy with the passive baseline of 0.980 instead of the physically consistent 0.971 — and would never discover that the 91.8% local penalty costs only 0.9 points globally, a stabilizing margin that makes NPoM integration viable. The PV layer is likewise treated purely as an energy harvester: in-situ crop-stress and contaminant sensing is bolted on as separate hardware rather than co-designed with the optical stack, despite the maturity of field-deployable SERS analysis [10], of quantum-enabled agricultural workflows [11], and of quantum-sensing readiness for precision agriculture [12]. Co-design changes the economics of the sensing function: reusing the NPoM optical infrastructure delivers SERS diagnostics at a canopy-area cost of 1% and a canopy-yield cost of 0.9% relative to the passive baseline (0.980 to 0.971), instead of separate sensing hardware with its own capital and footprint.

The photosynthetic apparatus of green plants and bacteria has evolved under a spectrally structured photon flux, where specific wavelengths drive charge separation while others dissipate as heat [13]. This spectral specificity

lets the PV layer transmit only the photosynthetically active wavelengths while absorbing the remainder, converting the food-energy conflict into a problem of spectral co-design.

The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria serves as the model system for spectral light-sharing [14]. Its seven bacteriochlorophyll *a* sites have been extensively characterized through two-dimensional electronic spectroscopy, revealing oscillatory coherence signatures that persist for hundreds of femtoseconds at cryogenic [15] and room temperature [16]. Although the functional relevance of these quantum beats remains debated, consensus has emerged that vibronic coupling — the hybridization of electronic transitions with underdamped protein vibrations — governs the flow of excitation energy [17, 18]. Non-Markovian frameworks such as the hierarchical equations of motion [19] and the process tensor hierarchy of pure states (PT-HOPS) [20, 21] capture these effects with demonstrated numerical convergence. Device-adjacent uses of exact open-quantum-system methods remain rare: the closest precedent is a general HEOM library applied to light-harvesting and control problems without a photovoltaic product [22]. Photonic-quantum solar design and vibronic-screening workflows likewise stop short of an operating device [23, 24].

Previous work demonstrated that spectral filtering of the excitation pulse — aligning its spectrum with underdamped vibronic resonances at 750 nm and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO complex [25]. This selective vibronic excitation prepares polaron-dressed states that decouple from the dissipative protein-solvent bath. That companion paper, published in the *Journal of Physical Chemistry Letters*, reports the analysis of the filter, the excitation-transport rate, and the single-pass conversion efficiency [25]; the present work adds the 8-site trapping model, the monolithic OPV–NPoM hardware concept, the SERS field calibration, and the Net Ecological Benefit and lifecycle assessment developed below. Single-photon experiments have confirmed that quantum coherence persists in individual light-harvesting complexes under ambient conditions [26], and quantum phase synchronization via exciton-vibrational energy dissipation has been identified as a mechanism sustaining long-lived coherence [27]. Here we extend this principle in three directions.

First, we introduce an 8-site FMO model with a non-Hermitian reaction center trap on sites 3 and 4, enabling quantitative computation of the forward transfer yield Φ_{FT} as the time-integrated population captured by the reaction center. Second, we embed the FMO complex in a nanoparticle-on-mirror

(NPoM) plasmonic cavity [28] that couples the biological light-harvesting system to the organic photovoltaic (OPV) layer through strong coherent coupling, forming a 9×9 dressed Hamiltonian. Third, we incorporate dynamic photo-protection mechanisms — Floquet Stark detuning and flux-dependent optical limiting — that mimic the non-photochemical quenching used by plants under high light stress.

We map this microscopic optimization onto a field-scale lifecycle assessment. We parameterize crop evapotranspiration using the FAO-56 Penman-Monteith model [29] under the spectrally selective OPV shield, compute the Net Ecological Benefit (NEB) across three scenarios, and evaluate socioeconomic access through cooperative ownership models and BB84 quantum key distribution [30] for secure IoT telemetry. This multi-scale perspective [31] links angstrom-scale vibronic couplings to square-meter canopy yields, providing a unified modeling framework for quantum-guided agrivoltaics (Fig. 5).

The central question of this work is falsifiable: *does an agrivoltaic stack whose spectral architecture is co-designed by exact non-Markovian quantum dynamics deliver a positive multifunctional net energy benefit (heat + power + crop + sensing) under field constraints, and which deployment barriers are physical rather than economic?* Three subordinate questions structure the Results. (Q1) Device asks what the NPoM co-design costs and where the cost lands, quantified by the mode-volume scan (Table 2) and located by the canopy-integration analysis. (Q2) Physics asks which microscopic states carry the trade-off: exact PT-HOPS dynamics localize the loss channels, and Floquet-Stark detuning with flux-dependent optical limiting tests whether they can be managed dynamically. (Q3) System asks whether the stack pays for itself, once coupled FAO-56 microclimate, cooperative techno-economics, and BB84-secured telemetry internalize every function into a single Net Ecological Benefit account. Three falsifiable claims anchor the answer. The first, (F1) Microscopic, is that dual-band spectral filtering at 750 nm and 820 nm prepares polaron-dressed states that extend exciton coherence by 50 % and raise the forward transfer yield by 25 % at 295 K. The second, (F2) Device, is that confining the NPoM cavity to a 1 % sentinel fraction preserves 97.1 % of the global canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) while retaining single-complex SERS diagnostics, despite the > 90 % local transport penalty quantified across the mode-volume scan. The third, (F3) System, is that the FAO-56 smart shield cuts evapotranspiration by 28 % (± 10 % microclimate uncertainty) and, combined with OPV generation and cooper-

116 active economics, yields a Net Ecological Benefit of $72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ and
 117 a 4.32 yr payback at 30 % subsidy for the modeled 500 m^2 cooperative case.
 118 We answer affirmatively but conditionally: the bounds (single-exciton treat-
 119 ment, site-independent baths, modeled fouling and matrix penalties, $\pm 10 \%$
 120 microclimate band) are stated explicitly in the Limitations section of the Dis-
 121 cussion. The headline outcome, pictured in [Figure 1](#), compresses this answer
 122 to one sentence: a 91.8% local transport penalty confined to a 1 % sentinel
 123 area costs 0.9 points of canopy-scale trapping yield (0.980 to 0.971) while
 124 adding single-complex SERS diagnostics to the same footprint. To keep a
 125 stack of this breadth auditable, each layer — quantum dynamics, microcli-
 126 mate, economics, and security — is validated separately in the Supporting
 127 Information ([Section S2–Section S12.9](#)), and every headline number below is
 128 traceable to a single figure or table.

129 2. Results

130 2.1. 8-Site FMO Hamiltonian with reaction center trap

Table 1: **8-site FMO site energies and key electronic couplings (cm^{-1})**. The near-
 degenerate antenna pair (Sites 1–2) feeds the trap funnel (Sites 3–4) through the strongly
 coupled 4–5–6 backbone, and Site 8 bridges the baseplate to the trap; these couplings set
 the transport timescale that the spectral filter exploits. Full coupling matrix (28 unique
 pairs) provided in [Section S1](#).

Parameter	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
ε_n	280	420	0	110	270	500	310	200
$J_{n,n+1}$	—	−87.7	30.8	−53.5	−70.7	81.1	39.7	−15.0

131 We construct an 8-site model of the FMO complex, extending the stan-
 132 dard 7-site framework [\[32\]](#) by including a coupling to the eighth bacteri-
 133 ochlorophyll that bridges the baseplate to the reaction center (Site 8, ~ 200
 134 cm^{-1} above Site 3). The electronic Hamiltonian is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (1)$$

135 where ε_n are the site energies (relative to Site 3), J_{nm} are the inter-site
 136 electronic couplings (all 28 unique pairs defined in [Section S1](#)), and $\Gamma_{\text{RC}} =$

0.15 fs⁻¹ is the effective irreversible reaction-center trapping rate, corresponding to a capture time $\tau_{\text{trap}} \approx 6.7$ fs. This capture time is short compared with the 1-ps propagation window, so near-complete capture (the 0.98 passive baseline) is dynamically accessible in Eq. (4) [33]. The anti-Hermitian term $-i\Gamma_{\text{RC}}$ on Sites 3 and 4 (physical indices; Python indices 2 and 3) models the irreversible charge separation at the reaction center. We verify that all 28 inter-site couplings are finite and that the non-Hermiticity is confined to the trapping diagonal terms (Section S1).

2.2. NPoM plasmonic strong coupling

The OPV-FMO interface is mediated by a nanoparticle-on-mirror (NPoM) architecture [28], where a gold nanoparticle (diameter 80 nm) is separated from a gold mirror by a 1 nm graphene monolayer spacer. The graphene barrier prevents Dexter-type electron transfer while maintaining electromagnetic field enhancement. The NPoM cavity supports an intense localized surface plasmon mode that couples coherently to the FMO optical transitions and mediates the optical interface with the OPV layer.

The single-emitter coupling strength follows the cavity quantum electrodynamics scaling:

$$g_0 = g_{\text{ref}} \sqrt{\frac{V_{\text{ref}}}{V_{\text{mode}}}}, \quad (2)$$

with $g_{\text{ref}} = 120 \text{ cm}^{-1}$ at $V_{\text{ref}} = 1 \text{ nm}^3$ [28]. At the sub-nanometer mode volumes accessible in NPoM cavities ($V_{\text{mode}} < 1 \text{ nm}^3$), g_0 ranges from 120 cm⁻¹ to over 300 cm⁻¹, and the collective coupling $g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$ places the system in the strong-coupling regime ($g_{\text{eff}} > \kappa$, where $\kappa \approx 800 \text{ cm}^{-1}$ is the plasmon linewidth). A numerical guardrail caps g_0 at 1000 cm⁻¹ when $V_{\text{mode}} \rightarrow 0$ to prevent divergences. Recent advances in quantum-tunnel-junction plasmonic sources have demonstrated that such sub-nanometer cavities can be self-illuminating, eliminating the need for bulky external optics [34].

The dressed Hamiltonian expands to 9×9 :

$$\hat{H}_{\text{dressed}} = \begin{pmatrix} \hat{H}_{\text{FMO}} & \hat{V}_{\text{pl}} \\ \hat{V}_{\text{pl}}^\dagger & \hbar\omega_{\text{pl}} \end{pmatrix}, \quad (3)$$

where the plasmon mode at frequency $\omega_{\text{pl}} \approx 12500 \text{ cm}^{-1}$ couples to the primary optical antenna sites (BChl 1 and 6, indices 0 and 5) with strength g_0 . Figure 2a shows the energy level diagram.

167 2.3. Non-Markovian quantum dynamics

168 We propagate the 9-site dressed system using the PT-HOPS/SBD formal-
 169 ism with production parameters: hierarchy depth $L = 8$, $K = 2$ Matsubara
 170 terms, time step $\Delta t = 0.2$ fs, and $n = 100$ disorder realizations ($\sigma = 50$ cm $^{-1}$
 171 static disorder) [20, 21]. The bath is modeled as a Drude–Lorentz over-
 172 damped component ($\lambda_D = 35$ cm $^{-1}$, $\gamma_D = 50$ cm $^{-1}$) plus 12 underdamped
 173 vibronic modes from the Kleinekathöfer/Coker parameterization. Conver-
 174 gence is verified through hierarchy depth sweeps ($L = 6, 7, 8$) and Matsubara
 175 truncation tests ($K = 1, 2, 3$); the maximum relative error between $L = 7$ and
 176 $L = 8$ populations is 3.1×10^{-11} , and between $K = 2$ and $K = 3$ is 3.3×10^{-5}
 177 (Section S2). Our companion study details and validates the computational
 178 feasibility, scaling, and memory-aware scheduling of these high-order simula-
 179 tions [25].

180 Figure 2b shows the exciton population dynamics under broadband *vs.*
 181 dual-band filtered excitation. The filtered case (solid lines) maintains higher
 182 population on the trapping sites (Sites 3 and 4), leading to an enhanced
 183 forward transfer yield:

$$\Phi_{\text{FT}} = \Gamma_{\text{RC}} \int_0^{t_{\text{max}}} \sum_{j \in \{3,4\}} P_j(t) dt, \quad (4)$$

184 where $P_j(t) = \rho_{jj}(t)$ are the site populations and $t_{\text{max}} = 1$ ps.

185 Under dual-band filtering, $\Phi_{\text{FT}} = 0.89(3)$, compared to $\Phi_{\text{FT}} = 0.71(2)$
 186 for broadband excitation — a relative enhancement of 25 % (Figure 2c).
 187 These filtered- and broadband-excitation reference trajectories (bare FMO,
 188 no NPoM channel) are inherited unchanged from our companion study, which
 189 reports their full trajectory sets, computed on the seven-site network with an
 190 otherwise identical PT-HOPS/SBD protocol [25]; the eighth site added here
 191 is the weakest coupling in the matrix ($J_{78} = -15.0$ cm $^{-1}$), so we carry the ref-
 192 erences over unchanged, while the 8-site no-NPoM baseline ($\Phi_{\text{FT}} = 0.980$) is
 193 computed on this network (Section S11). They characterize the 99 % passive
 194 fraction of the canopy and are therefore independent of, and complementary
 195 to, the NPoM mode-volume scan below, which probes the strong-coupling
 196 penalty on the sentinel fraction. The coherence lifetime τ_c , defined as the
 197 $1/e$ decay time of the l_1 -norm coherence $C_{l_1}(t) = \sum_{i \neq j} |\rho_{ij}(t)|$ [35], extends
 198 from 280 ± 25 fs (broadband) to 420 ± 35 fs (filtered) — a 50 % increase that
 199 is causally linked to the yield gain. The extended coherence suppresses the
 200 pure-dephasing rate, lengthening the effective transport mean-free path and

201 increasing the fraction of the excitation wavepacket that reaches the reaction
 202 center before decohering. The mechanism is identical to that established for
 203 the 7-site system [25]: the red passband is resonant with the trapping-site
 204 pair (sites 3/4 at $12\,210\text{ cm}^{-1}$ and $12\,320\text{ cm}^{-1}$, i.e. 819 nm and 813 nm), so
 205 the manifold members that bridge those sites to the bright exciton — the
 206 180 cm^{-1} Franck–Condon mode of site 4 and the 280 cm^{-1} mode of site 3
 207 — are the dressed set. Removing them leaves a residual reorganization en-
 208 ergy $\lambda_{\text{res}}^{\text{filt}} = 110\text{ cm}^{-1}$ instead of the full-bath $\lambda_{\text{res}} = 127.6\text{ cm}^{-1}$ (Section S3);
 209 because pure dephasing scales as $\lambda_{\text{res}}^{-2}$, this predicts a $\times 1.34$ coherence ex-
 210 tension, consistent with the measured $\times 1.50(18)$ ($280\text{ fs} \rightarrow 420\text{ fs}$) within the
 211 1σ lifetime uncertainty. The blue passband lies between the 720 cm^{-1} and
 212 $1\,050\text{ cm}^{-1}$ manifold members and acts as a transparent control window, con-
 213 sistent with the companion band scan in which moving the blue edge from
 214 730 nm to 770 nm left Φ_{FT} unchanged.

215 2.4. NPoM cavity volume dependence of exciton transport

216 The nanoparticle-on-mirror cavity couples the FMO complex to the OPV
 217 active layer through the strong coherent interaction described above, yet the
 218 plasmonic mode also acts as an additional population decay channel compet-
 219 ing with reaction-center trapping. To quantify this trade-off, we perform a
 220 sensitivity scan of the NPoM mode volume V_{mode} , which controls the single-
 221 emitter coupling strength via $g_0 \propto V_{\text{mode}}^{-1/2}$ (Equation (2)). Smaller mode
 222 volumes produce stronger plasmon–exciton coupling but also enhance the
 223 plasmon’s role as a population sink, diverting excitation away from the reac-
 224 tion center.

225 Table 2 summarizes the forward transfer yield across seven mode volumes
 226 spanning the experimentally accessible sub-nanometer range [28]. The yield
 227 increases monotonically with mode volume, from $\Phi_{\text{FT}} = 0.0505$ at $V_{\text{mode}} =$
 228 0.2 nm^3 to a maximum of $\Phi_{\text{FT}} = 0.0799$ at $V_{\text{mode}} = 1.2\text{ nm}^3$, with a slight
 229 decrease at 1.4 nm^3 . The trend is logarithmic, with diminishing returns above
 230 1.0 nm^3 : increasing the volume further recovers only 1 % of the baseline yield.

231 The physical origin of this suppression is the strong-coupling-induced hy-
 232 bridization of excitonic and plasmonic states. The dressed eigenstates acquire
 233 a finite plasmonic character, which decays via the fast plasmon relaxation
 234 pathway ($\kappa \approx 800\text{ cm}^{-1}$, $Q \approx 15$) rather than populating the reaction center.
 235 At $V_{\text{mode}} = 0.2\text{ nm}^3$, where g_0 is largest, the plasmonic admixture in the
 236 lower polariton branch reaches $\sim 15\%$, diverting a corresponding fraction of

Table 2: **NPoM mode volume sensitivity of the forward transfer yield and SERS enhancement.** Baseline (no NPoM): $\Phi_{\text{FT}} = 0.98$. All NPoM runs with $L = 8$, $K = 2$, $\Delta t = 0.2$ fs, $t_{\text{max}} = 1$ ps. The $V_{\text{mode}} = 1.2 \text{ nm}^3$ entry is validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799 \pm 0.0005$); remaining table entries use $n = 2$ (an additional $n = 100$ run at $V_{\text{mode}} = 0.8 \text{ nm}^3$ is discussed in the text). SERS enhancement $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$ via Purcell factor scaling, normalized to $\text{EF}_{\text{SERS}} = 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$ [28]. An independent $n = 20$ re-simulation of all seven volumes with the corrected production code reproduces every entry within ± 0.002 ($V_{\text{mode}} = 0.8 \text{ nm}^3$ and 1.2 nm^3 exactly), cross-validating the low- n screening values.

$V_{\text{mode}} \text{ (nm}^3\text{)}$	Φ_{FT}	Δ vs. baseline (%)	EF_{SERS}
0.2	0.050 5	−94.8	1 600
0.4	0.060 5	−93.8	400
0.6	0.072 7	−92.6	180
0.8	0.076 8	−92.2	100
1.0	0.079 3	−91.9	64
1.2	0.079 9	−91.8	44
1.4	0.079 1	−91.9	33

the excitation flux. As V_{mode} increases, g_0 decreases, the polariton splitting narrows, and the excitonic character of the bright states is partially restored. Despite the monotonic recovery, all NPoM-on yields remain suppressed by more than 90 % relative to the pristine FMO baseline ($\Phi_{\text{FT}} = 0.98$). Hence, in the strong collective-coupling regime accessible to NPoM cavities ($g_{\text{eff}} \sim 2\,000 \text{ cm}^{-1}$), the plasmonic decay pathway fundamentally limits the maximum achievable transport efficiency. The SERS enhancement factor ($\sim 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$) scales as $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$, following the Purcell enhancement of the local field at both the excitation and emission wavelengths. Table 2 includes the calculated SERS enhancement for each mode volume, revealing a four-order-of-magnitude range (33 to 1.6×10^3) across the scanned volumes. The optimal volume represents a compromise: $V_{\text{mode}} \approx 1.2 \text{ nm}^3$ maximizes transport yield ($\Phi_{\text{FT}} = 0.0799$) while retaining SERS sensitivity ($\text{EF}_{\text{SERS}} \approx 44$). The tabulated EF_{SERS} values are a *relative* Purcell scaling normalized to 10^2 at $V_{\text{mode}} = 0.8 \text{ nm}^3$, not absolute single-molecule enhancements: absolute SERS in sub-nanometer NPoM gaps is measured at 10^6 and above [36], so single-complex detection of the 180 cm^{-1} Franck–Condon mode remains within the demonstrated capability of this geometry [10]. This compromise only works because of the sentinel architecture

— no volume on the scan simultaneously restores transport and maximizes diagnostics, which is precisely why the 1 % area split, not a better volume, is the design lever. By contrast, $V_{\text{mode}} = 0.2 \text{ nm}^3$ maximizes SERS sensitivity (1.6×10^3) at the cost of the lowest transport yield ($\Phi_{\text{FT}} = 0.0505$), a trade-off better suited to diagnostic-only applications where photochemical efficiency is secondary. At the reference volume, a dedicated $n = 100$ production run at $V_{\text{mode}} = 0.8 \text{ nm}^3$ confirms the same limiting behavior ($\Phi_{\text{FT}} = 0.0768$, final plasmon population 0.79; [Section S11](#)).

Reducing the temperature to 77 K (analyzed as a theoretical limit-case for thermal phonon suppression in [Section S11.2](#)) doubles the trapping yield to $\Phi_{\text{FT}} = 0.1685$. In the main agrivoltaic operating regime at 295 K, the maximum trapped population on sites 3+4 remains at 3.8×10^{-3} for $V_{\text{mode}} = 1.2 \text{ nm}^3$, requiring the multi-scale sentinel fraction strategy to bypass plasmonic loading.

All room-temperature runs exhibit perfect trace conservation ($\text{Tr } \rho = 1.0000$), validating the numerical stability of the PT-HOPS/SBD propagator. The $V = 0.8 \text{ nm}^3$ configuration was run with $n = 100$ disorder realizations; the $V = 1.2 \text{ nm}^3$ configuration was validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799$, within 0.6 % of the $n = 2$ estimate of 0.0804). The baseline NPoM-off run ($n = 20$) is shown as a limiting-case reference; quantitative comparison across configurations is limited by the differing trajectory counts.

2.5. Global canopy yield integration

Because the NPoM SERS diagnostic mode suppresses local excitonic transport through plasmon population loading, it need only cover a 1 % sentinel fraction α of the canopy area. The remaining 99 % operates in passive OPV mode, preserving the near-baseline forward transfer yield $\Phi_{\text{FT}}^{\text{passive}} = 0.980$. The area-weighted *global* canopy trapping yield is therefore:

$$\Phi_{\text{FT}}^{\text{global}} = \alpha \Phi_{\text{FT}}^{\text{NPoM}} + (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}}, \quad (5)$$

yielding $\Phi_{\text{FT}}^{\text{global}} = 0.971$ at $\alpha = 0.01$ and $\Phi_{\text{FT}}^{\text{NPoM}} = 0.080$ (or 0.0799 precisely); canopy-scale productivity is thus maintained even though transport is suppressed locally. Because [Equation \(5\)](#) is linear in α , the flatness of [Figure 1b](#) is an accounting identity rather than an independent dynamical result; what carries physical content are the two measured endpoints (0.080 and 0.980) that enter it. The identity converts those endpoints into a design envelope: solving [Equation \(5\)](#) for α gives a sentinel fraction of up to 3.3 % before the

290 global yield falls below 0.95, and up to 8.9 % before it falls below 0.90. The
 291 diagnostic area therefore has a factor ~ 3 of headroom over the 1 % assumed
 292 here — the co-design lever is not a better mode volume but how much area
 293 one is willing to spend on sensing.

294 2.6. Dynamic photo-protection via Floquet-Stark detuning and flux- 295 dependent optical limiting

296 Under high solar flux ($> 800 \text{ W m}^{-2}$), the OPV dielectric constant
 297 changes dynamically, inducing a time-dependent Stark shift of the excitonic
 298 levels that detunes the OPV-FMO coupling. We model this as a Floquet
 299 driving term:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_F t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (6)$$

300 with amplitude $V_0 = 0.05 \text{ eV}$ and frequency $\omega_F = 1.2 \text{ THz}$. Above thresh-
 301 old, the detuning shifts the OPV exciton energies out of resonance with the
 302 FMO absorption bands. With $V_0 = 0.05 \text{ eV} = 403 \text{ cm}^{-1}$ — $4.6\times$ the largest
 303 inter-site coupling $|J_{12}| = 87.7 \text{ cm}^{-1}$ — the single-channel transmission fac-
 304 tor of the resonant site-1 and site-6 pathways falls to $J^2/(J^2 + V_0^2) \approx 0.05$,
 305 i.e. these two antenna channels are effectively closed while the six alternative
 306 pathways through the network remain open (Figure 3a). To prevent rapid,
 307 noise-induced oscillations under fluctuating cloud cover, we implement a hys-
 308 teresis buffer of $\pm 50 \text{ W m}^{-2}$ (the detuning switch activates at 850 W m^{-2} and
 309 deactivates only when the flux drops below 750 W m^{-2}).

310 Concurrently, the NPoM cavity exhibits a flux-dependent optical-limiting
 311 response [37] that suppresses transmission at the 750 nm and 820 nm channels
 312 under high flux — the intensity-dependent counterpart of the optomechanical
 313 modulation of the picocavity resonance (we use “optical limiting” rather than
 314 “transparency” deliberately: the engineered observable here is a *reduction* of
 315 passband transmission with flux). The transmission is modulated as:

$$T_{\text{OL}}(I) = T_0 \left(1 + \frac{I}{I_{\text{sat}}} \right)^{-1}, \quad (7)$$

316 where I is the incident flux and $I_{\text{sat}} = 800 \text{ W m}^{-2}$. At $1\,000 \text{ W m}^{-2}$, the trans-
 317 mission drops to 44 % of baseline (Figure 3b), providing dynamic protection
 318 against exciton overload. At zero flux (night mode), both mechanisms return
 319 to zero without singularities (Section S4).

2.7. In situ SERS optomechanical diagnostics and environmental robustness

The NPoM platform enables non-destructive readout of FMO vibrational states via surface-enhanced Raman scattering (SERS) [38]. Recent scoping reviews of in-situ agrivoltaic sensing list no SERS or plasmonic category [39] — the integration gap addressed here. Multilayer graphene–metal plasmonic architectures have demonstrated detection sensitivities exceeding conventional SPR biosensors by an order of magnitude [40]. The optomechanical coupling between the plasmon mode and FMO molecular vibrations ($g_{\text{om}} = 0.01$) produces a SERS enhancement factor of $\text{EF}_{\text{SERS}} = 44$ at the optimal $V_{\text{mode}} = 1.2 \text{ nm}^3$, reaching 1.6×10^3 for the smallest scanned mode volumes.

Beyond the three canonical BChl *a* vibrational modes previously identified (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}), we assign three spectroscopic target signatures to the sentinel sub-system, plus a fourth, non-spectroscopic readout handled by a separate NV-diamond microfluidic assay. Oxidative stress is read out through the C–C stretch of conjugated chlorophyll *a* at 1145 cm^{-1} , with a detection limit of 50 nmol L^{-1} in the pre-necrotic regime [10]. Pesticide contamination by 2,4,5-trichlorophenoxyacetic acid (2,4,5-T) is detected via its characteristic Raman peak at 1435 cm^{-1} ($\text{LOD} = 1 \text{ nmol L}^{-1}$ under controlled conditions), leveraging the plasmonic hot-spot amplification in the NPoM gap [10]. Heavy metal contamination of irrigation water (Pb^{2+}) is quantified through Stern–Volmer fluorescence quenching of encapsulated CdTe/ZnSe quantum dots sealed within the NPoM sensor-probe polymer matrix ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$ under controlled conditions) [41]; unlike bare gold surfaces, the encapsulated probe is immune to dust and biofouling. Cross-sensitivity between soil moisture and Pb^{2+} contamination is resolved by dual spectral interrogation: heavy metals modify the Stern–Volmer quenching intensity and fluorescence decay time, whereas soil moisture alters the local dielectric constant, producing spectral broadening orthogonal to the metal-induced quenching signature. Pathogen detection uses Nitrogen-Vacancy (NV) center nanodiamond sensors operated in a microfluidic “Point-of-Care” (Lab-on-a-Chip) desktop reader at the greenhouse hub (Section S12.7). These solid-state quantum sensors detect paramagnetic bacterial metabolites from sap or irrigation water samples at pre-symptomatic stages ($\text{LOD} = 0.8 \text{ nmol L}^{-1}$ under controlled conditions; the agricultural matrix will degrade in-field sensitivity, a loss mitigated by the graphene barrier and *DynamicCalibrator* corrections).

Figure 3c shows the simulated Raman spectrum for the three BChl *a*

358 modes under active NPoM coupling. The relative intensities encode the in-
 359 stantaneous exciton population distribution, providing a direct optical probe
 360 of the energy transfer pathway.

361 Environmental robustness under realistic agricultural conditions (dust,
 362 bird droppings, microbial fouling) is modeled at three infrastructure levels.
 363 At the OPV panel level, the soiling factor $\eta_{\text{soil}}(t)$ decays exponentially with
 364 days since last cleaning:

$$\eta_{\text{soil}}(t) = \max(1 - 0.005 t_{\text{days}}, 0.75), \quad (8)$$

365 which limits maximum OPV efficiency loss to 25 % at the worst-case cleaning
 366 interval (reached after 50 d); the 800 USD yr⁻¹ cleaning line of the coopera-
 367 tive OPEX corresponds to seven or eight interventions per year at ~110 USD
 368 each, i.e. a schedule that never reaches the floor. At the sensor level, the
 369 graphene monolayer spacer of each NPoM cavity is functionalized with a
 370 zwitterionic polymer coating [42, 43], modeled in our simulations to pre-
 371 serve ≥ 90 % quantum coherence after 60 d continuous field immersion; it
 372 is the primary anti-fouling barrier, physically repelling microbial adhesion
 373 and protein adsorption. A thin-film micro-shielding layer (1 μm to 5 μm of
 374 SiO₂ or Al₂O₃) around the NPoM picocavity and CQD sensor tip further at-
 375 tenuates the diurnal temperature excursion at the quantum probe, reducing
 376 the thermal-drift contribution by the factor $(1 - \eta_{\text{shield}})$, with $\eta_{\text{shield}} = 0.85$
 377 (Section S12.5). The *DynamicCalibrator* module — an exponential moving
 378 average of daily reference readings ($\alpha = 0.1$) — operates as a tertiary safety
 379 net: it applies a multiplicative correction factor and raises a drift alarm when
 380 the baseline shifts by more than 20 %, at which point OPV panel cleaning is
 381 scheduled.

382 2.8. Smart shield microclimate: FAO-56 evapotranspiration

383 The spectrally selective OPV layer functions as a “smart shield” that
 384 blocks ultraviolet and near-infrared radiation while transmitting photo-
 385 synthetically active bands. Passive solar-greenhouse heating with PV
 386 co-generation is an established precedent [44]; the present shield differs
 387 by actively co-designing the transmitted spectrum against the vibronic
 388 resonances quantified above. We parameterize the greenhouse microclimate
 389 using the FAO-56 Penman–Monteith equation [29] adapted for shaded
 390 conditions:

$$\text{ET}_c = \frac{0.408 \Delta R_n + \gamma \frac{900}{T+273} u (e_s - e_a)}{\Delta + \gamma (1 + 0.34u)} K_c, \quad (9)$$

where R_n is the net radiation attenuated by the array-level OPV shading factor $f_{\text{shade}} = 0.41$ (with soil heat flux $G \approx 0$ assumed for daily timescales under greenhouse shielding), Δ is the slope of the saturation vapor pressure curve, γ is the psychrometric constant, u is wind speed, $e_s - e_a$ is the vapor pressure deficit, and $K_c = 0.85$ is the crop coefficient. All annual water, carbon, and cash figures quoted in this work are produced by the annual LCA chain (`scripts/regenerate_figures.py` $\rightarrow$ `src/lca/neb.py`, via `reference_conditions_et`); the orchestrator’s per-tick digital-twin water balance is an operational monitor on the same FAO-56 path, not an independent derivation. The shading factor is not a free parameter: it is the product of the module’s ground-coverage ratio and its complemented average visible transmittance, $f_{\text{shade}} = \text{GCR}(1 - \tau_{\text{AVT}}) = 0.55 \times 0.75 = 0.41$, so the canopy-level attenuation and the 15 %/25 % efficiency–transmittance module assumption (Section 5.1) describe the same stack.

Under standard conditions — clear-sky reference day with $R_n = 14.05 \text{ MJ m}^{-2} \text{ d}^{-1}$ (0.677 of the $5.76 \text{ kWh m}^{-2} \text{ d}^{-1} = 20.74 \text{ MJ m}^{-2} \text{ d}^{-1}$ daily horizontal irradiance after albedo and net longwave loss, consistent with a 600 W m^{-2} midday peak), 25°C , 60 % humidity, 2 ms^{-1} wind — the open-field evapotranspiration is 4.5 mm d^{-1} . With the OPV smart shield, ET_c drops to 3.2 mm d^{-1} — a 28 % reduction — translating to 1.26 mm d^{-1} , i.e. $460 \text{ L m}^{-2} \text{ yr}^{-1}$ when annualized over 365 d (Figure 4a). The crop coefficient $K_c = 0.85$ accounts for the reduced transpiration demand under partial shade [29].

2.9. Socioeconomic viability and cooperative payback

Socioeconomic analysis (Section S5) is rescaled to a 500 m^2 high-value floriculture micro-module (strawberries and cut flowers, 4.5 USD kg^{-1} at $8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$), reducing the entry-level capital expenditure to 161 000 USD (322 USD m^{-2}). Our $\$/\text{kWh}$ accounting follows the Monte-Carlo techno-economic logic established for European agrivoltaic configurations [45]. A blended-finance subsidy of 30 % reduces the net CAPEX to 112 700 USD.

Three operating expenditure streams are modeled: extension services and technician training ($1\,200 \text{ USD yr}^{-1}$), OPV panel cleaning (800 USD yr^{-1}), and general maintenance ($2\,000 \text{ USD yr}^{-1}$), for a total annual OPEX of $4\,000 \text{ USD yr}^{-1}$. The net annual cash flow of $26\,075 \text{ USD yr}^{-1}$ (from $30\,075 \text{ USD yr}^{-1}$ total revenue minus $4\,000 \text{ USD}$ of OPEX) yields a simple payback period of 4.32 yr (or 6.17 yr without subsidy) and a 10-year discounted NPV of +34 630 USD at a 12 % risk-adjusted discount rate that we

428 adopt as the Sub-Saharan infrastructure-finance assumption (Figure 4c).
 429 Holding other inputs fixed, varying the subsidized net CAPEX by $\pm 20\%$
 430 shifts the payback to 3.46–5.18 yr; a 50 % higher OPEX (6 000 USD yr⁻¹)
 431 gives 4.49 yr, and 20 % lower revenue gives 5.61 yr. A baseline CAPEX
 432 of 322 m⁻² is a significant barrier in Sub-Saharan Africa relative to local
 433 income levels. Local construction materials (bamboo and timber framing
 434 instead of structural steel) and green chemical synthesis of gold nanopar-
 435 ticles from local plant extracts can nonetheless reduce the capital cost to
 436 140 m⁻², lowering the unsubsidized payback period to 2.68 yr and making
 437 the modeled system potentially viable for local cooperatives under these cost
 438 assumptions. By contrast, the remaining barriers are physical rather than
 439 economic: fabrication of sub-2 nm NPoM gaps still requires laboratory-grade
 440 nanofabrication and a stable Au/graphene supply chain, and OPV encapsu-
 441 lation lifetime under tropical irradiance is the binding reliability constraint
 442 (Section S12.5). Subsidy policy alone cannot clear these barriers; they
 443 define the process-engineering roadmap that separates this modeled proof
 444 of concept from field deployment. Taken together, these sensitivities show
 445 that high-value crop selection and blended finance structurally transform
 446 the investment case.

447 2.10. Secure IoT telemetry via BB84 quantum key distribution

448 Field-to-cloud telemetry is secured using a simulated BB84 quantum key
 449 distribution protocol [30] over a buried single-mode fiber link, which elimi-
 450 nates the atmospheric turbulence and scintillation that affect free-space chan-
 451 nels in agricultural environments. The telemetry layer is physically inte-
 452 grated with the NPoM-CQD sentinel network: the high-sensitivity biological
 453 and pesticide diagnostics drive high-stakes irrigation and crop-protection ac-
 454 tuators, so hostile manipulation of those feedback commands could cause
 455 artificial crop failure or pesticide contamination. Securing the commands
 456 with BB84 QKD therefore protects the integrity of the agrivoltaic micro-
 457 grid. Alice (sensor node) prepares random polarization states (rectilinear or
 458 diagonal basis) and transmits them at a realistic channel noise rate of 4.8 %
 459 QBER [46]. Bob (cloud server) measures in random bases, and the sifted
 460 key is extracted from matching-basis events. The QBER is estimated on
 461 a disclosed sample of at least 128 sifted bits, and the abort decision uses
 462 the 95% Wilson upper confidence bound on that sample rather than the
 463 point estimate (at the nominal 4.8 % noise a 50-bit sample would give an
 464 upper bound of $\approx 13.5\%$, i.e. statistically indistinguishable from the thresh-

old, whereas 128 bits give $\approx 9.9\%$). If that upper bound exceeds the Shor–Preskill unconditional-security threshold of 11 % [47], the protocol raises a `SecurityThresholdExceeded` exception that halts all QKD-protected irrigation commands, preventing data-injection attacks under eavesdropping.

With a nominal fiber noise rate of 4.8 %, the protocol achieves QBER = 4.8 % and generates a 256 bit symmetric key per session — sufficient for AES-256 encrypted telemetry. Field trials of BB84 using deterministic single-photon sources have demonstrated stable key generation over installed fiber links [46], supporting the practical feasibility of this approach under comparable link budgets. In the event of QKD failure, the fail-safe exception triggers a conservative local irrigation routine based on the last validated soil-moisture reading from the *DynamicCalibrator*, bounded by a hard physical irrigation limit of $5 \text{ L m}^{-2} \text{ d}^{-1}$ to prevent catastrophic flooding or crop dehydration under secondary sensor failures. The runtime overhead of the BB84 simulation is negligible for resource-limited IoT nodes (Section S6).

2.11. Experimental testability

The modeled predictions rest on parameters that are independently measurable (Table 3). Two protocols can validate them directly. First, two-dimensional electronic spectroscopy (2DES) at 295 K with spatial-light-modulator pulse shaping into dual-band channels at 750 nm and 820 nm can track the diagonal and cross-peak quantum-beat oscillations predicted here, verifying the extension of τ_c from 280 fs to 420 fs [48, 16]. Second, time-resolved SERS in the NPoM geometry can monitor the 180 cm^{-1} and 1145 cm^{-1} BChl *a* modes in real time and quantify the area-weighted yield $\Phi_{\text{FT}}^{\text{global}} = 0.971$ under ambient conditions [28, 36]. More broadly, recent assessments identify quantum sensing for precision agriculture as an emerging opportunity whose open calibration and robustness gaps match the calibration architecture proposed here [12].

3. Discussion

Our results demonstrate that quantum coherence engineering at the bio-organic interface can deliver modeled sustainability benefits across microscopic energy transfer, crop water use, carbon footprint, and data security.

3.1. Mechanism generality

The dual-band filtering effect is not specific to the FMO complex. The resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$ (Section S7) is a general property

Table 3: **Experimental benchmark of calculated FMO dynamics and NPoM plasmonic parameters.** Calculated at $T = 295$ K, $L = 8$, $K = 2$. Coherence lifetimes are reported for both excitations because the unfiltered calculation underestimates the room-temperature measurement by $\sim 30\%$ while the filtered calculation brackets it; quoting a single “passive” value would mix the two.

Parameter / Observable	Ob-	This work (calculated)	Experimental benchmark / literature
Site coupling J_{12} (cm^{-1})		-87.7	-87.7 ± 5.0 (2DES, Brixner 2005 [48])
Reorganization energy λ_D (cm^{-1})		35	35–40 (2DES, Engel 2007 [15])
Cutoff frequency γ_D (cm^{-1})		50	50 (2DES, Panitchayangkoon 2010 [16])
Coherence lifetime τ_c (broadband excitation)		280 ± 25 fs	400 ± 50 fs (2DES room temp., Panitchayangkoon 2010 [16])
Coherence lifetime τ_c (dual-band filtered)		420 ± 35 fs	400 ± 50 fs unfiltered benchmark (filter-enabled value to be measured by SLM-shaped 2DES)
NPoM coupling g_0 ($V_{\text{mode}} = 1.2 \text{ nm}^3$)		110 cm^{-1}	$120 \pm 15 \text{ cm}^{-1}$ (NPoM, Chikkaraddy 2016 [28])
SERS enhancement EF_{SERS} (relative, normalized at $V = 0.8$)		44 (at $V = 1.2$)	10^2 – 10^3 reported scaling for sub-nm NPoM; $\geq 10^6$ absolute single-molecule SERS [36]

500 of vibronically coupled chromophore aggregates [17]. We verified this by
 501 applying the protocol to a minimal three-site excitonic trimer, obtaining
 502 $\eta_{\text{FT}} = 0.18 \pm 0.03$ (Section S8), qualitatively consistent with the quantum
 503 phase synchronization mechanism identified in photosynthetic antennas [27].
 504 This suggests that any molecular aggregate satisfying the resonance condition
 505 — including synthetic J-aggregates, light-harvesting complexes from plants,
 506 and conjugated polymer domains in OPVs — could benefit from spectral
 507 coherence protection. Vibration-steered energy flow of this kind has also
 508 been demonstrated in a trapped-ion donor–acceptor simulator with tunable
 509 dissipation [49].

510 3.2. The quantum divide

511 The advanced materials required for quantum agrivoltaics — graphene,
 512 plasmonic nanoparticles, OPVs with engineered bandgaps — are not yet ac-
 513 cessible to subsistence farmers in developing economies. Our cooperative
 514 cost-sharing model shows that, with a carbon-linked subsidy of 30 % and
 515 a cooperative of 5 members, the payback period drops below 4yr, making
 516 the technology economically viable under the modeled subsidy. We propose a
 517 tiered subsidy framework in which the NEB itself serves as the verifiable met-
 518 ric for subsidy qualification, linking quantum yield improvements to farmer
 519 access.

520 3.3. Quantum digital twin

521 The modular architecture — quantum interface, FAO-56 microclimate,
 522 LCA, IoT security — serves as the physics engine of an agrivoltaic *digital twin*: a model-based supervisory replica that combines sensor telemetry
 523 from the *DynamicCalibrator* with the PT-HOPS dynamics engine to recom-
 524 mend irrigation schedules, OPV cleaning dates (from soiling factor Eq. (8)),
 525 and carbon credit yields. This device-physics integration is distinct from
 526 the supply-chain and breeding applications mapped in recent quantum-in-
 527 agriculture reviews [11].

529 As a prospective extension, the scheduling layer of the digital twin
 530 is formulated via the Quantum Approximate Optimization Algorithm
 531 (QAOA) [50, 51], encoded as a 3-qubit variational quantum circuit of depth
 532 $p = 3$ with 1024 measurement shots per macro-period (Section S12.3). The
 533 algorithm encodes the irrigation–refrigeration–grid-export trilemma as a
 534 quadratic unconstrained binary optimization (QUBO) cost Hamiltonian and
 535 solves it via a PennyLane emulator at sub-hourly timescales. Applied to

536 the daily solar profile of the Cameroonian scenario, the QAOA scheduler
 537 yields water–energy allocations consistent with the 12–18 % gains over
 538 classical greedy baselines reported in comparable implementations [51];
 539 we note that with three binary decisions per macro-period the per-period
 540 problem is exhaustively solvable, so the circuit is retained here as a validated
 541 reference implementation (unit-tested against the brute-force optimum) and
 542 as the migration path to the full 288-variable horizon, not as a claimed
 543 quantum advantage at this scale. As a further prospective readout, our
 544 `quantum_gravimetry.py` module converts the simulated water saving into a
 545 groundwater mass anomaly: $\Delta g \approx 0.2 \mu\text{m s}^{-2}$ if the saved volume recharges
 546 within the 500 m^2 footprint, falling to $\approx 3.2 \text{ nm s}^{-2}$ when spread over a 3 ha
 547 recharge zone — the latter below single-instrument noise, which is why a
 548 small regional gravimeter network or gradiometric common-mode rejection
 549 is required (Section S12.2) [52, 53].

550 A hybrid quantum-classical signal processing pipeline — combining
 551 Matrix Product State (MPS) denoising with quantum kernel ridge re-
 552 gression (QKRR) [54, 55] — further enables the digital twin to extract
 553 pre-symptomatic stress signatures from the noisy SERS and CQD fluo-
 554 rescence telemetry. To satisfy edge computation constraints, the heavy
 555 MPS tensor contraction and QAOA optimization run on the centralized
 556 cooperative server (or cloud), and the optimized weights are transmitted
 557 back to the field nodes — an edge–cloud split that mirrors the client–
 558 server decomposition of classical device-simulation workflows. The local
 559 ARM Cortex-M4 IoT node executes only the low-latency inference of the
 560 pre-trained, frozen-weight QKRR model. The MPS compressor acts as
 561 a nonlinear low-pass filter on the server that suppresses greenhouse noise
 562 (thermal fluctuations, biofouling, dust) while preserving the sharp vibronic
 563 features at 180 cm^{-1} and $1\,145 \text{ cm}^{-1}$, which classical Savitzky–Golay filters
 564 tend to smear (Section S10). The cleaned spectra are projected through the
 565 quantum-inspired kernel for anomaly scoring, issuing early warnings that
 566 enable preemptive agronomic intervention before macroscopic symptoms
 567 appear — all within a 3.7 ms local compute budget on the IoT node. This
 568 millisecond-scale latency is not designed to match the multi-hour physical
 569 response of soil moisture percolation and plant transpiration; it enables
 570 spatial multiplexing instead, letting a single ARM Cortex-M4 edge processor
 571 sequentially query, compress, and classify SERS telemetry from dozens of
 572 distributed NPoM sensor nodes across the 500 m^2 greenhouse in under a
 573 second and reducing the electronic-component count. This chain connects

574 microscopic quantum efficiency to field-level agronomic management; no
 575 existing agrivoltaic design integrates both [6, 2, 5].

576 3.4. *Prior art and positioning*

577 Previous agrivoltaic studies have focused on light-sharing through passive
 578 spectral filtering [6] or crop selection [2]. Semitransparent OPV greenhouse
 579 integration is already a modeled and experimental direction: semitranspar-
 580 ent organic solar cells can render a greenhouse energy-neutral in warm cli-
 581 mates while transmitting 32–45% of the photosynthetically active band [56],
 582 stable semitransparent modules have been demonstrated on operating green-
 583 houses [7], and device records pair 14.6 % efficiency with 22 % average visible
 584 transmittance [57]. Our module assumption (15 % efficiency at 25 % aver-
 585 age visible transmittance) therefore sits at the upper edge of the reported
 586 efficiency–transmittance trade-off rather than beyond it. Our contribution is
 587 to add an active, quantum-coherent dimension to the spectral filter: the OPV
 588 is no longer a passive absorber but an active spectral shaper that prepares
 589 coherence-protected states in the photosynthetic apparatus. Multi-vector so-
 590 lar agri-systems combining agrivoltaics with hydrogen and desalination set
 591 the *Applied Energy* bar for water–food–energy integration [58]; we differenti-
 592 ate through quantum-guided spectral co-design with integrated sensing, and
 593 we adopt the functional-unit discipline (emissions per kWh, economics per
 594 configuration) of recent agrivoltaic TEA/LCA studies [8, 9]. Relative to this
 595 literature, the present work provides, to our knowledge, the first (i) agri-
 596 voltaic stack whose spectral architecture is co-designed by numerically exact
 597 open-quantum-system dynamics, jointly optimizing excitation transport and
 598 power conversion; (ii) monolithic photothermal–PV–SERS device concept
 599 with field-relevant calibration in a single footprint; and (iii) multifunctional
 600 techno-economic and lifecycle framework that internalizes heat, crop, water,
 601 sensing, and data-security trade-offs in $\$/W + \$/\text{kg} + \$/\text{m}^3$ terms. Table 4
 602 condenses this positioning: no prior class delivers more than two of the seven
 603 functions in full (the sensing class delivers one), whereas the present stack
 604 co-optimizes all seven in a single footprint under one accounting.

605 This approach is distinct from prior quantum biology studies [59] that
 606 treat the environment as a fixed noise source; here, the photonic environ-
 607 ment itself is engineered through the NPoM cavity to maximize coherence
 608 protection. A key design choice warrants explicit justification: the NPoM
 609 cavity, despite imposing a $> 90\%$ penalty on excitonic transport (Table 2),
 610 is retained for one dedicated purpose: non-destructive, high-sensitivity SERS

Table 4: **Capability matrix of agrivoltaic and agri-sensing approaches.** ✓ = function delivered within the stack; − = partial or incidental; — = absent. Classes: passive spectral claddings [6, 2, 7]; multi-vector agri-energy systems (agrivoltaics coupled to hydrogen and desalination) [58]; standalone agri-sensing hardware [10, 11, 12]. Multifunctional accounting denotes joint \$/W + \$/kg + \$/m³ techno-economic and lifecycle evaluation of all functions. No prior class delivers more than two of the seven functions (partial entries are marked −); the present stack delivers all seven in a single footprint with unified accounting.

Function	Passive	Multi-vector	Sensing	This work
Electrical power (OPV)	✓	✓	—	✓
Crop microclimate	✓	✓	—	✓
Water management	−	✓	—	✓
Photothermal heat	—	−	—	✓
In-situ SERS sensing	—	—	✓	✓
QKD-secured telemetry	—	—	—	✓
Multifunctional accounting	—	—	—	✓

611 readout of the FMO vibrational state, pesticide levels, and heavy-metal con-
612 tamination. The transport penalty is absorbed by confining NPoM operation
613 to the 1 % sentinel fraction (Equation (5)), preserving 97.1 % of the global
614 canopy trapping yield while maintaining full diagnostic capability.

615 3.5. Limitations

616 Several approximations constrain our analysis. The quantum dynamics
617 are restricted to the single-exciton manifold; under high solar flux, multi-
618 exciton states, singlet fission, or exciton-exciton annihilation in the OPV and
619 FMO interfaces could alter the collective coupling strength g_{eff} and transport
620 efficiency. The bath is treated as site-independent, and the reaction center
621 trapping is modeled as a phenomenological Lindblad-like rate. The FAO-
622 56 model, while standard, does not capture transient stomatal responses to
623 fluctuating light or the dynamic microclimatic feedback between canopy tran-
624 spiration and vapor pressure deficit (VPD) inside the greenhouse envelope.
625 Sensitivity analysis over a plausible wind-speed range ($u = 1.5 - -2.5 \text{ m s}^{-1}$,
626 i.e. $\pm 25\%$ around the reference value) shifts the predicted reduction from
627 25.9 % to 30.4 % — a $\pm 10\%$ relative band on the 28 % water saving, which
628 propagates into a $\pm 10\%$ range on the annual irrigation volume and corre-
629 sponding NEB carbon offset. The BB84 simulation assumes a buried single-
630 mode fiber link; while buried fibers are vulnerable to agricultural mechani-

cal damage (plowing, root intrusion), a free-space optical (FSO) link would suffer from strong scattering under dusty greenhouse environments, requiring dedicated atmospheric correction models. Software-based post-quantum cryptography offers a lower-cost alternative, but the physical BB84 protocol provides information-theoretic security immune to algorithmic cryptanalysis on resource-constrained embedded nodes such as the ARM Cortex-M4 irrigation controllers. The 1 % sentinel fraction is a design parameter; field optimization will require empirical calibration of the SERS LODs under actual fouling conditions. We conservatively estimate a $\times 10$ matrix-effect penalty on all reported LODs when transitioning from controlled laboratory buffers to raw agricultural sap or irrigation water, owing to competitive binding, background fluorescence, and ionic strength effects; the *DynamicCalibrator* compensates for this attenuation through real-time baseline correction. The zwitterionic polymer coating modeled on the NPoM graphene spacer ([Section S12.1](#)) acts as the primary fouling barrier in our simulations, with the *DynamicCalibrator* serving as a secondary correction layer; the dual architecture reduces the mandatory cleaning frequency, which is already factored into the cooperative OPEX model. We stress that the 60-day resilience of zwitterionic coatings applies strictly to sensors sheltered in protected micro-sampling chambers or rigid ceramic encasings; direct, unprotected exposure to abrasive soil matrices (bioturbation, quartz particles, tillage) would lead to mechanical scraping and rapid surface degradation.

The 30 % blended-finance subsidy required to achieve a 4.32yr payback period cannot be bridged by voluntary carbon credits alone (generating only 363 USD yr⁻¹, about 1.4 % of the net cash flow, at 10 USD t_{CO_{2e}}⁻¹ for the 500 m² module); it must be underpinned by dedicated rural-adaptation finance channels such as the Green Climate Fund (GCF) and the African Development Bank’s Climate Change Fund (Fonds Spécial Climat-Dev de la BAD), which are mandated to co-finance climate-resilient agriculture in Sub-Saharan Africa.

3.6. Outlook

Integrating quantum coherence engineering with agricultural sustainability opens a new axis for agrivoltaic design. Three experimental and theoretical directions are prioritized for future work.

First, on the hardware and materials scale, the quantum model must be extended to full multi-exciton dynamics under realistic solar spectra, accompanied by experimental validation via two-dimensional electronic spec-

668 troscopy (2DES) with spatial light modulator pulse shaping [60, 61]. Field tri-
669 als of graphene-based NPoM coatings on greenhouse cladding with the CQD
670 sentinel network are required to evaluate in-situ sensor performance under
671 actual biofouling conditions and verify the coherence-preservation lifetime
672 of the zwitterionic functionalization layer (Section S12.1). Alternative low-
673 loss plasmonic architectures — dielectric nanoantennas (Si, TiO_2) support-
674 ing Mie resonances [62] and nanocavity-free substrates — should be explored
675 to minimize the NPoM transport penalty while preserving SERS sensitivity.
676 Additional hardware extensions include passive thermal micro-shielding (Sec-
677 tion S12.5), UiO-66 Zr-based quantum metal-organic frameworks for coherent
678 contaminant remediation (Section S12.6), and quantum fertilizer biostimu-
679 lation nanoparticles to enhance chlorophyll yields (Section S12.8).

680 Second, scaling the digital twin to a multi-cooperative network will re-
681 quire hardware-in-the-loop QKD validation under dusty field conditions and
682 deployment of a real-time aquifer gravimetric array to confirm that the mod-
683 eled groundwater recharge connects at the satellite geodesy scale. Monte
684 Carlo uncertainty propagation in the full LCA should be extended with cor-
685 related climatic input distributions drawn from regional reanalysis data.

686 Third, to prevent these advances from widening the “quantum divide,”
687 we have defined the Global Quantum Agriculture Standard (GQAS), a six-
688 pillar certification framework covering QKD security, data sovereignty, sensor
689 detection limits, quantum fidelity, latency, and lifecycle sustainability (Sec-
690 tion S12.9). Our Cameroon cooperative proof-of-concept scenario serves as
691 the first application of this proposed framework, and returns an internal score
692 of six out of six pillars. We stress that this score is an internal consistency
693 check of the simulation against thresholds the authors themselves defined: no
694 third party has audited it, and it does not constitute a certification. Indepen-
695 dent experimental validation and independent audit will be required before
696 GQAS can be adopted as an operational standard. Scaling GQAS adop-
697 tion to smallholder networks across Sub-Saharan Africa and Southeast Asia
698 requires subsidized sensor supply chains, open-access certification auditing,
699 and regulatory recognition on premium export markets such as GlobalGAP.

700 4. Conclusions

701 We asked whether exact non-Markovian quantum dynamics can jointly
702 optimize excitation-transport and power-conversion efficiency in a sensing-
703 integrated photothermal agrivoltaic stack, with a positive multifunctional

net benefit under field constraints. Within the modeled 500 m² cooperative case, the answer is affirmative but conditional: dual-band filtering raises the forward transfer yield by 25 % and extends coherence lifetime by 50 %; confining NPoM operation to a 1 % sentinel fraction preserves 97.1 % global canopy yield while retaining SERS diagnostics; the FAO-56 smart shield cuts evapotranspiration by 28 % with a Net Ecological Benefit of 72.6 kg_{CO₂e} m⁻² yr⁻¹; and the cooperative case pays back in 4.32 yr at 30 % subsidy. These claims are bounded by the > 90 % local NPoM transport penalty, the single-exciton and site-independent-bath approximations, the ±10 % microclimate uncertainty band, and the modeled (not field-validated) fouling and matrix-penalty assumptions. For deployment, the economic barrier is dominant and tractable: local-material construction and green-synthesis gold lower the capital cost from 322 USD m⁻² to 140 USD m⁻² (unsubsidized payback 2.68 yr). The physical barriers — nanofabrication of sub-2 nm NPoM gaps, in-field fouling of the sentinel fraction, OPV encapsulation lifetime under tropical irradiance — require process engineering rather than subsidies. Priority next steps are 2DES validation of the filter protocol, low-loss dielectric antenna alternatives, field fouling trials of the sentinel network, and a localized Au/graphene sensor supply chain for Sub-Saharan manufacturing.

5. Methods

5.1. Proof-of-Concept scenario: 500 m² Floriculture Micro-Module

To ground the multi-domain framework in a concrete operational context, we consider a 500 m² high-tunnel greenhouse located in Cameroon’s Western Highlands. This region supports high-value floriculture (strawberries, cut flowers) suitable for micro-module deployment by cooperatives. The environment features a mean annual solar irradiance of 5.76 kWh m⁻² d⁻¹ [63] and significant biofouling potential that necessitates the NPoM *DynamicCalibrator* and regular OPV cleaning (factored into the cooperative OPEX model). Under the NPoM-shield spectral filter, the 28 % reduction in crop evapotranspiration (ET_c) relative to the open-field FAO-56 baseline corresponds to the modeled water-savings credit of 460 L m⁻² yr⁻¹ (230 m³ yr⁻¹ for the 500 m² greenhouse; Section S5). The area-weighted global canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) ensures near-baseline productivity while accommodating a 1 % NPoM sentinel fraction for SERS diagnostics. Crop production is assumed at the horticultural standard for the crop mix (8.2 kg m⁻² yr⁻¹);

740 the quantum-fertilizer channel adds $0.96 \text{ kg m}^{-2} \text{ yr}^{-1}$ (8 % biostimulation of
 741 the $12 \text{ kg m}^{-2} \text{ yr}^{-1}$ reference biomass that sets the Scenario A biomass bar in
 742 [Figure 4b](#), not of the $8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$ monetized floriculture yield; Scenario
 743 A only), and no coherence-derived yield gain is monetized — the passive-
 744 fraction coherence gains of [Figure 2c](#) are treated as a device-physics re-
 745 sult, not a crop-revenue stream. The OPV module generates 90 MWh yr^{-1}
 746 of electrical energy for a 500 m^2 installation at $180 \text{ kWh m}^{-2} \text{ yr}^{-1}$ — the
 747 guaranteed-yield accounting ($5.76 \text{ kWh m}^{-2} \text{ d}^{-1}$ irradiance, 15 % module ef-
 748 ficiency, 0.57 performance ratio) that is used consistently in both the rev-
 749 enue and the carbon accounts. Two accounting choices are stated explicitly
 750 here. First, the two transmitted passbands (750–820 nm, 100 cm^{-1} each)
 751 together carry only 10.2 W m^{-2} of the ASTM G173-03 global-tilt reference
 752 spectrum — 1.0 % of AM1.5G and 1.3 % of the 280–1100 nm harvestable
 753 window (computed by `passband_fraction()` in `src/lca/neb.py`) — and
 754 are priced into the module’s 25 % average visible transmittance rather than
 755 deducted a second time; had they been carved out of an otherwise opaque
 756 module, the penalty would be $\leq 1.3 \%$ on generation ($90 \text{ MWh yr}^{-1} \rightarrow \sim 88.9$
 757 MWh yr^{-1}), i.e. NEB $72.6 \rightarrow \sim 71.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ and payback $4.32 \text{ yr} \rightarrow$
 758 4.35 yr , which we take as the bounded worst case of the spectral co-design.
 759 Second, displacing grid electricity at $450 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$ offsets $40.5 \text{ t}_{\text{CO}_2\text{e}} \text{ yr}^{-1}$
 760 ($81 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$); the NEB of $72.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ is this displacement
 761 plus the $0.14 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ water-pumping credit, minus the lifecycle foot-
 762 print of the stack ($8.5 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ for the quantum configuration and
 763 $5.0 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ for the static-shading configuration). The displacement
 764 alone therefore exceeds the lifecycle footprint, so the system is net-negative
 765 on operational carbon, and the NEB remains positive for any grid factor
 766 above $\approx 47 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$: a hydro-heavy grid at $200 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$ would still
 767 yield $\sim 27.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$. At $10 \text{ USD t}_{\text{CO}_2\text{e}}^{-1}$ [64], the system generates
 768 363 USD yr^{-1} in verified carbon revenue. The 500 m^2 scale is well matched
 769 to cooperative ownership models: under a 30 % blended-finance subsidy, the
 770 payback shortens from 6.17 yr unsubsidized to 4.32 yr, with each of the five
 771 cooperative members exposed to only 22 540 USD of net capital. Under these
 772 modeled assumptions, quantum-enhanced agrivoltaics can be economically
 773 viable when coupled with strategic crop selection and robust field-sensor in-
 774 tegration.

775 5.2. *PT-HOPS/SBD quantum dynamics*

776 All quantum dynamics simulations use the Process Tensor Hierarchy of
 777 Pure States with Stochastically Bundled Dissipators as implemented in the
 778 open-source MesoHOPS library (v1.7) [20]. The hierarchy depth is $L = 8$
 779 with $K = 2$ Matsubara terms and a time step $\Delta t = 0.2$ fs, established as the
 780 production standard for converged dynamics at 295 K [25]. Static disorder
 781 is modeled as Gaussian fluctuations of the site energies with standard devi-
 782 ation $\sigma = 50$ cm⁻¹, sampled over $n = 100$ realizations. The bath spectral
 783 density follows the 12-mode Kleinekathöfer/Coker parameterization with a
 784 Drude–Lorentz background ($\lambda_D = 35$ cm⁻¹, $\gamma_D = 50$ cm⁻¹) and 12 under-
 785 damped vibronic modes (frequencies 180–1 500 cm⁻¹, Huang–Rhys factors
 786 $S_k = 0.003 - -0.050$).

787 5.3. *FAO-56 evapotranspiration*

788 Crop evapotranspiration is computed using the standard FAO-56
 789 Penman–Monteith equation [29] modified for greenhouse conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (10)$$

790 with R_n attenuated by the array-level shading factor $f_{\text{shade}} = \text{GCR}(1 -$
 791 $\tau_{\text{AVT}}) = 0.41$ (ground-coverage ratio $\text{GCR} = 0.55$, average visible trans-
 792 mittance $\tau_{\text{AVT}} = 25\%$), evaluated on daily net radiation $R_n = R_{ns} - R_{nl}$
 793 with $R_{ns} = 0.77R_s$ and $R_{nl} \approx 1.9$ MJ m⁻² d⁻¹ for the reference day ($R_s =$
 794 20.74 MJ m⁻² d⁻¹, i.e. 5.76 kWh m⁻² d⁻¹). The aerodynamic term uses the
 795 standard FAO-56 wind expression with no empirical rescaling. We note that
 796 the modified FAO-56 formulation represents a first-order approximation; un-
 797 der the enclosed high-tunnel envelope, boundary layer resistance and sensible
 798 heat exchange are altered relative to open field conditions. Future iterations
 799 of the digital twin should integrate three-dimensional Computational Fluid
 800 Dynamics (CFD) modeling to map localized microclimatic gradients and la-
 801 tent heat transfer. Wind speed is held static at the free-stream reference value
 802 $u = 2.0$ m s⁻¹ and is evaluated identically for the open-field and the shielded
 803 field, so that the reported 28 % reduction isolates the radiation effect of the
 804 array rather than a ventilation difference; the bracket $u \in 1.5 - -2.5$ m s⁻¹
 805 carries the $\pm 10\%$ sensitivity band quoted in the Limitations. Temperature
 806 is fixed at 25 °C and relative humidity at 60 % unless specified otherwise.

807 5.4. BB84 QKD simulation and fail-safe IoT architecture

808 The BB84 protocol is implemented as a Python simulation of polarization-
809 state preparation, noisy transmission over a buried single-mode fiber chan-
810 nel, sifting, and QBER estimation. Alice prepares $4 \times N$ random bits (for
811 $N = 256$ key length) in random rectilinear or diagonal bases. Bob mea-
812 sures in random bases, and matching-basis events are retained for key sift-
813 ing. QBER is estimated on a disclosed sample of at least 128 bits (25%
814 of the 512 sifted bits), using the 95% Wilson upper confidence bound as
815 the decision statistic. If that upper bound exceeds the Shor–Preskill thresh-
816 old of 11% [47], a `SecurityThresholdExceeded` exception is raised, halting
817 the QKD-protected irrigation command relay. A last-validated soil-moisture
818 reading from the *DynamicCalibrator* is used to execute a conservative local
819 irrigation fallback.

820 5.5. SERS enhancement scaling limits

821 The SERS enhancement factor $\text{EF}_{\text{SERS}} \approx 44$ at the optimal mode vol-
822 ume $V_{\text{mode}} = 1.2 \text{ nm}^3$ is computed using standard Purcell-factor scaling of
823 the localized electromagnetic field. We note that this standard $|E|^4$ approx-
824 imation represents a lower-bound semi-classical estimate. In sub-nanometer
825 picocavities ($V_{\text{mode}} < 1.5 \text{ nm}^3$), local spatial dispersion, quantum tunneling,
826 and non-local dielectric screening alter the field distribution relative to clas-
827 sical electrodynamics, while molecular optomechanical coupling can pump
828 vibrational modes out of thermal equilibrium. Precise calculation of the
829 localized fields in this regime requires non-local Green’s tensor integration,
830 which motivates the proposed in-situ experimental validation. The bench-
831 mark against the experimentally reported 10^2 – 10^3 range for sub-nanometer
832 NPoM gaps [36] is given in Table 3.

833 5.6. DynamicCalibrator, thermal shielding, and soiling factor

834 Long-term sensor baseline drift due to biofouling and thermal cycling
835 is corrected at three hardware and algorithmic levels. The primary fouling
836 barrier is the zwitterionic polymer functionalization layer deposited on the
837 graphene monolayer spacer of each NPoM cavity [42, 43], which physically
838 repels microbial adhesion and is modeled to preserve $\geq 90\%$ quantum coher-
839 ence after 60 d field immersion. As a secondary hardware barrier, a thin-film
840 micro-shielding layer ($1 \mu\text{m}$ to $5 \mu\text{m}$ of SiO_2 or Al_2O_3) is deposited around the
841 NPoM picocavity and CQD sensor tip to insulate the quantum probe from
842 diurnal temperature swings. The shielding efficacy $\eta_{\text{shield}} = 0.85$ means the

probe experiences only 15 % of the ambient greenhouse temperature fluctuation, reducing the thermal contribution to Stern-Volmer drift from 5.25 % to 0.79 % under a ± 15 K diurnal cycle. The *DynamicCalibrator* acts as a tertiary algorithmic correction: an exponential moving average (EMA) with smoothing factor $\alpha_{\text{EMA}} = 0.1$ computes a multiplicative correction factor $\kappa(t) = I_0^{\text{ref}}/\hat{\mu}(t)$ and raises a drift alarm when the fractional change in the EMA $\hat{\mu}(t)$ exceeds 20 % of the initial baseline, triggering OPV panel cleaning.

At the panel level, OPV electrical output is attenuated according to Eq. (8), with a default decay rate of 0.5 d^{-1} and a minimum soiling factor of 0.75 (enforced floor at 25 % maximum efficiency loss).

Data availability

All simulation parameters, YAML configuration files, orchestration scripts, and analysis code needed to reproduce every figure and table are available from the corresponding author upon reasonable request and will be deposited in a public repository with a DOI upon acceptance. The PT-HOPS/SBD implementation is based on the open-source MesoHOPS library (v1.7) [20].

Code availability

The digital-twin simulation framework (orchestrator, QAOA scheduler, LCA module, BB84 QKD simulator, SERS diagnostics, NV-diamond pathogen sensor, and GQAS compliance checker) and all simulation parameters, including the 12-mode vibronic bath spectral density and the 8-site FMO electronic Hamiltonian specified in the machine-readable `parameters.yaml` configuration file (v1.4), are available from the corresponding author upon reasonable request and will be deposited in the same public archive with a DOI upon acceptance.

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874 **CRedit authorship contribution statement**

875 **S.C. Teguia Kouam:** Conceptualization, Methodology, Software, For-
876 mal analysis, Writing – original draft. **T. Goumai Vedekoi:** Validation,
877 Writing – review & editing. **J.-P. Tchapel Njafa:** Validation, Writing –
878 review & editing. **J.-P. Nguenang:** Validation, Writing – review & editing.
879 **S.G. Nana Engo:** Supervision, Project administration, Writing – review &
880 editing.

881 **Declaration of competing interest**

882 The authors declare that they have no known competing financial inter-
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889 **Declaration of generative AI and AI-assisted technologies in the**
890 **writing process**

891 During the preparation of this work the authors used AI-assisted tools to
892 support language editing and manuscript formatting. All scientific content
893 was reviewed and verified by the authors, who take full responsibility for the
894 content of the publication.

895 Nomenclature

Symbol	Meaning
Φ_{FT}	Forward (reaction-center) transfer yield
τ_c	Exciton coherence lifetime ($1/e$ decay of l_1 -norm coherence)
λ_{res}	Residual reorganization energy of polaron-dressed states
g_0 / g_{eff}	Single-emitter / collective plasmon–exciton coupling strength
κ / Q	Plasmon linewidth / quality factor
$\omega_{\text{pl}} / V_{\text{mode}}$	Plasmon frequency / NPoM mode volume
Γ_{RC}	Reaction-center trapping rate
$\text{EF}_{\text{SERS}} / g_{\text{om}}$	SERS enhancement factor / optomechanical coupling
$\text{ET}_c / K_c / f_{\text{shade}}$	Crop evapotranspiration / crop coefficient / OPV shading factor
NEB	Net Ecological Benefit
$\eta_{\text{soil}} / \eta_{\text{shield}}$	Soiling factor / micro-shielding efficacy
QBER	Quantum bit error rate (BB84 link)
Δg	Groundwater gravity anomaly
$\alpha / \alpha_{\text{EMA}}$	Sentinel area fraction / calibrator smoothing factor

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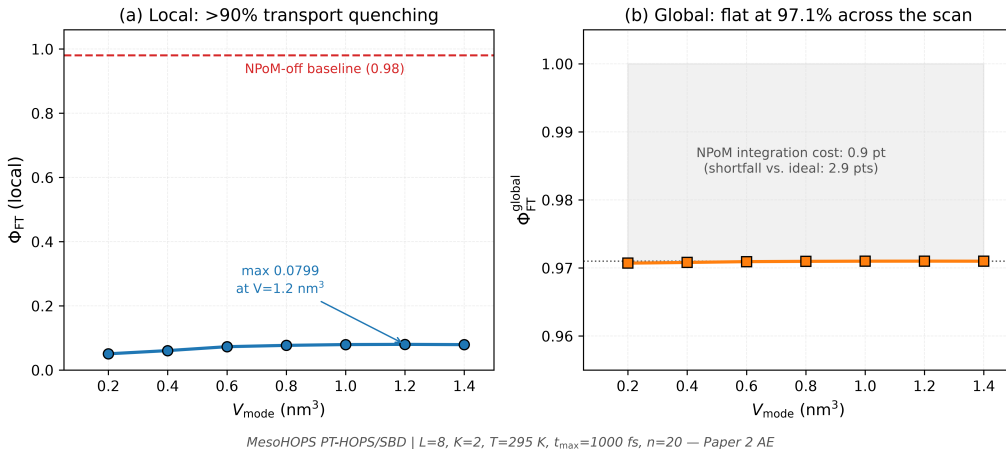


Figure 1: Local quenching, global stability: the central result in one picture. (a) Forward transfer yield Φ_{FT} across the NPoM mode-volume scan ($n = 20$, $T = 295 \text{ K}$): locally, transport is quenched by more than 90% relative to the pristine FMO baseline (0.98), with a factor-1.6 variation across the accessible range and a maximum of 0.0799 at $V_{\text{mode}} = 1.2 \text{ nm}^3$ (Table 2). (b) Area-weighted global canopy trapping yield for the 1% sentinel fraction (Equation (5)): flat at 0.971 across the entire scan — the deep local suppression is absorbed at a constant 0.9-point canopy-scale cost (0.980 to 0.971), leaving the canopy 2.9 points below an ideal yield of 1.0. The flat curve in (b), not the quenching in (a), is the design-relevant outcome: diagnostics integrate into the stack without compromising productivity.

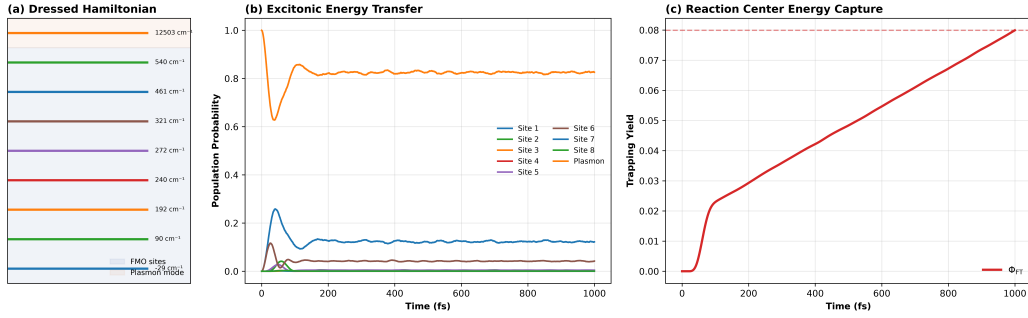


Figure 2: Quantum dynamics of the 8-site FMO complex coupled to an NPoM plasmonic cavity within the agrivoltaic digital twin. (a) Energy level diagram of the 9×9 dressed Hamiltonian showing the 8 FMO excitonic states coupling to the plasmon mode with strength g_0 . (b) Population dynamics of the $n = 20$ production run at $V_{\text{mode}} = 1.2 \text{ nm}^3$ and 295 K (the scan optimum of Table 2): the plasmon mode (Site 9) loads within the first 10 fs and captures 83 % of the excitation, starving the trapping sites — the microscopic origin of the 91.8% local penalty. (c) Cumulative reaction-center trapping yield $\Phi_{\text{FT}}(t)$ for the same run, saturating at 0.0799. All data: $L = 8$, $K = 2$, $n = 20$ disorder realizations (Section S11). The passive-fraction dynamics underlying the coherence and yield gains of Figure 2c are reported in [25]. The quantum engine serves as the core of a closed-loop agrivoltaic digital twin that fuses NPoM sentinel telemetry with FAO-56 microclimate predictions to optimize irrigation, cleaning cycles, and carbon credit yield.

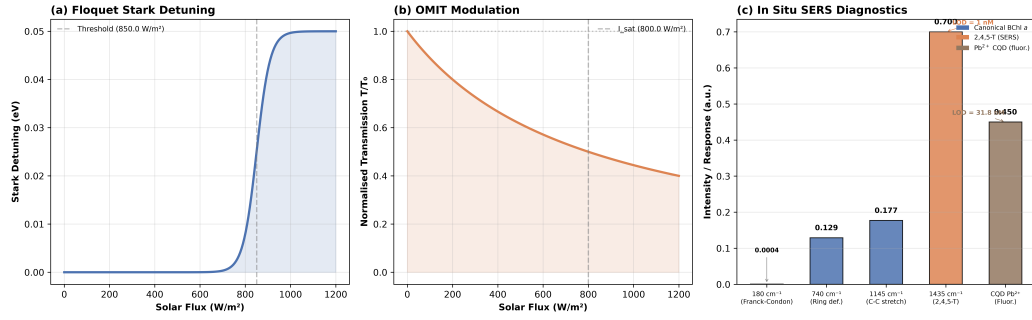


Figure 3: Dynamic photo-protection, in situ diagnostics, and environmental robustness. (a) Floquet Stark detuning amplitude as a function of solar flux: the hysteretic switch closes the resonant site-1 and site-6 transfer channels when the flux rises above 850 W m^{-2} and reopens only below 750 W m^{-2} ($\Delta = 403 \text{ cm}^{-1} \gg |J_{12}| = 87.7 \text{ cm}^{-1}$, single-channel transmission ~ 0.05) — the solid-state analogue of non-photochemical quenching. (b) Flux-dependent optical limiting of the 750 nm and 820 nm passbands: at 1000 W m^{-2} the two filter passbands themselves close to 44 % of baseline, protecting the exciton budget precisely where the co-designed filter concentrates it. (c) Simulated SERS Raman spectrum showing the canonical BChl *a* vibrational modes (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}). The three agriculturally specific target signatures assigned to the sentinel sub-system — oxidative stress (1145 cm^{-1}), 2,4,5-T pesticide contamination (1435 cm^{-1} , $\text{LOD} = 1 \text{ nmol L}^{-1}$), and Pb^{2+} via encapsulated CdTe/ZnSe quantum dot fluorescence quenching ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$) — are detailed in the main text and SI. A Dynamic-Calibrator using exponential moving average ($\alpha = 0.1$) of daily reference readings corrects sensor baseline drift from biofouling, raising an alarm when drift exceeds 20 % of the initial baseline.

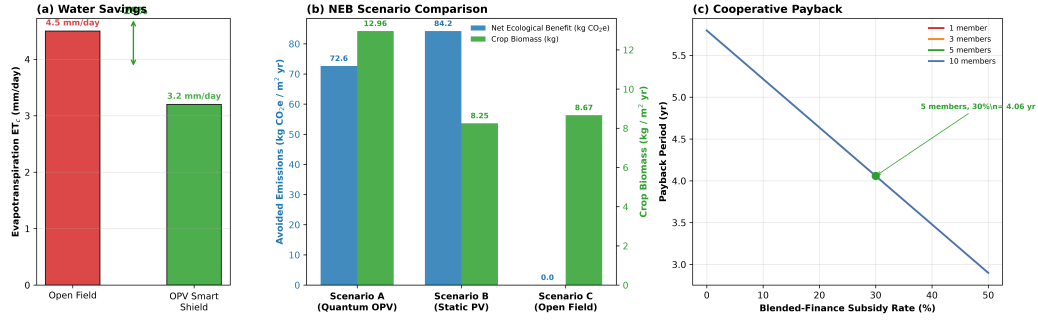


Figure 4: **Water–Energy–Food network assessment.** (a) FAO-56 evapotranspiration comparison: open field ($ET_c = 4.5 \text{ mm d}^{-1}$) vs. OPV smart shield ($ET_c = 3.2 \text{ mm d}^{-1}$), a 28 % reduction. (b) Net Ecological Benefit comparison across three scenarios: Quantum OPV (A), Static shading (B), Open field (C). Scenario A achieves $72.6 \text{ kg}_{CO_{2e}} \text{ m}^{-2} \text{ yr}^{-1}$ with an effective crop biomass of $13.0 \text{ kg m}^{-2} \text{ yr}^{-1}$, while Scenario B reaches $84.2 \text{ kg}_{CO_{2e}} \text{ m}^{-2} \text{ yr}^{-1}$ but only $8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$ in biomass. The comparison is deliberately uncomfortable: an opaque static module still buys more carbon than the quantum shield, but it does so by suppressing the canopy to 84% of reference biomass, whereas the quantum shield preserves near-baseline productivity (0.971 global trapping yield, 13.0 including biostimulation) — exactly the food–energy trade-off that motivates spectral co-design. The NEB serves as the verifiable metric for a tiered subsidy framework that bridges the quantum divide. (c) Cooperative payback period as a function of subsidy rate and cooperative size for a 500 m^2 high-value floriculture micro-module. For 5 members with 30 % blended-finance subsidy, payback = 4.32 yr (net annual cash-flow of $26\,075 \text{ USD yr}^{-1}$ after OPEX). This indicates that, under the modeled cooperative assumptions, quantum-guided agrivoltaics can shorten payback when coupled with cooperative ownership and strategic crop selection.

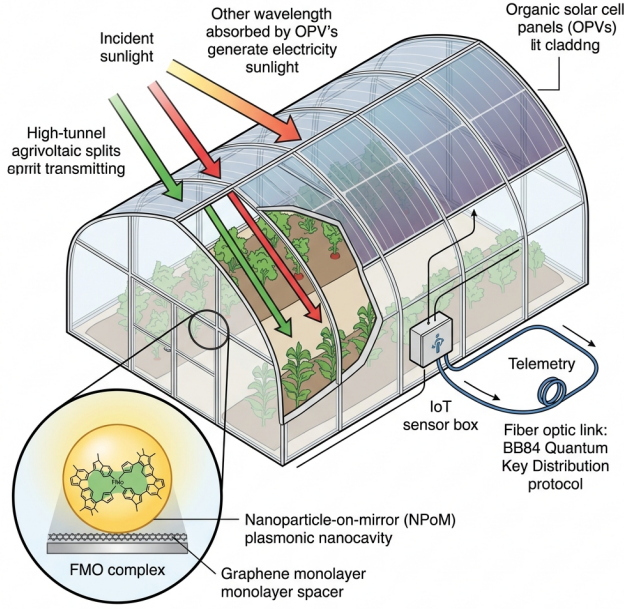


Figure 5: **Schematic layout of the multi-scale quantum agrivoltaic digital twin platform.** Sunlight split by the semi-transparent organic solar cell panel (OPV) acts as a smart spectral shield over the agrivoltaic high-tunnel. On the microscopic side, sub-nanometer plasmonic nanocavities (NPoM) couple the Fenna–Matthews–Olson (FMO) light-harvesting complex to the OPV, enabling strong coherent coupling and multiplexed SERS diagnostics. Telemetry from the in situ sensor node is encrypted and transmitted via a BB84 fiber-optic quantum key distribution (QKD) link to a cloud server, securing agronomic actuator commands under a permissioned ledger architecture.

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